Resistance

A resistor conducts electricity but resists the flow of current.

Resistance is measured in ohms (Ω).

Nichrome metal has a high resistance to electricity.

It is used to convert electrical energy into heat energy.

The resistance of a substance depends upon three factors:

1. .....
2. .....
3. .....

Resistance Questions

Measure the resistance of:

1 Electronic resistor .....

2 Range of dimmer ..... to .....

3 Unknown resistor .....

4 300 mm of thick nichrome wire

5 150 mm of thick nichrome wire

6 300 mm of thin nichrome wire

7 150 mm of thin nichrome wire

8 300 mm of copper wire

9 150 mm of copper wire

Resistance Questions

10 What happens to the resistance as the thickness of the resistor changes ?

11 What happens to the resistance of the resistor as the length changes ?

12 Does the type of substance a conductor is made of effect the resistance of a conductor ?

13 Explain your answer to Q12.

14 What other appliances besides heaters use resistors in their operation ? List three.